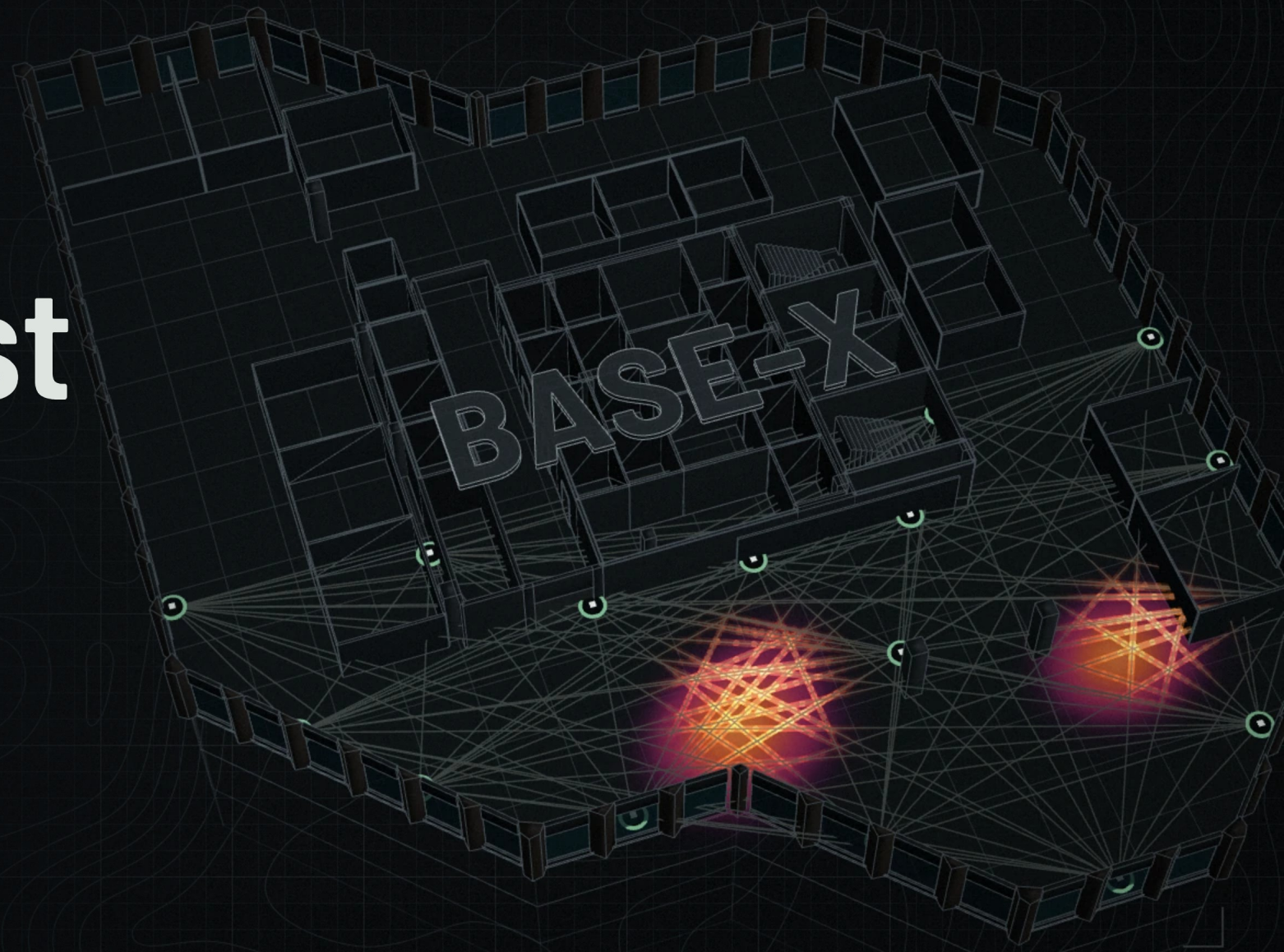
CODE METAL FORGE 2026 · TEAM 12

Shadowcast

Never go in blind.

Drop WiFi pucks around a building.
See who's moving inside, through walls, live.



NICOLAS · SIMON · ANDREW · SIV

BASE-X · 60 STATE STREET · FLOOR 14

Before anyone goes in, they need to know who’s moving inside

SPECIAL OPERATIONS

Is anyone in this room or building, and where?

That answer shapes how a team makes entry.

SEARCH AND RESCUE

Where are the people we can’t see?

At a collapse, fire or disaster, it tells responders where to search first.

OPERATING ENVIRONMENT

Behind walls, in smoke and dark, without reliable GPS, cell or fixed comms. Cameras and thermal stop at the wall; handheld radar sees one wall at a time.

Who we build for, and what we assume

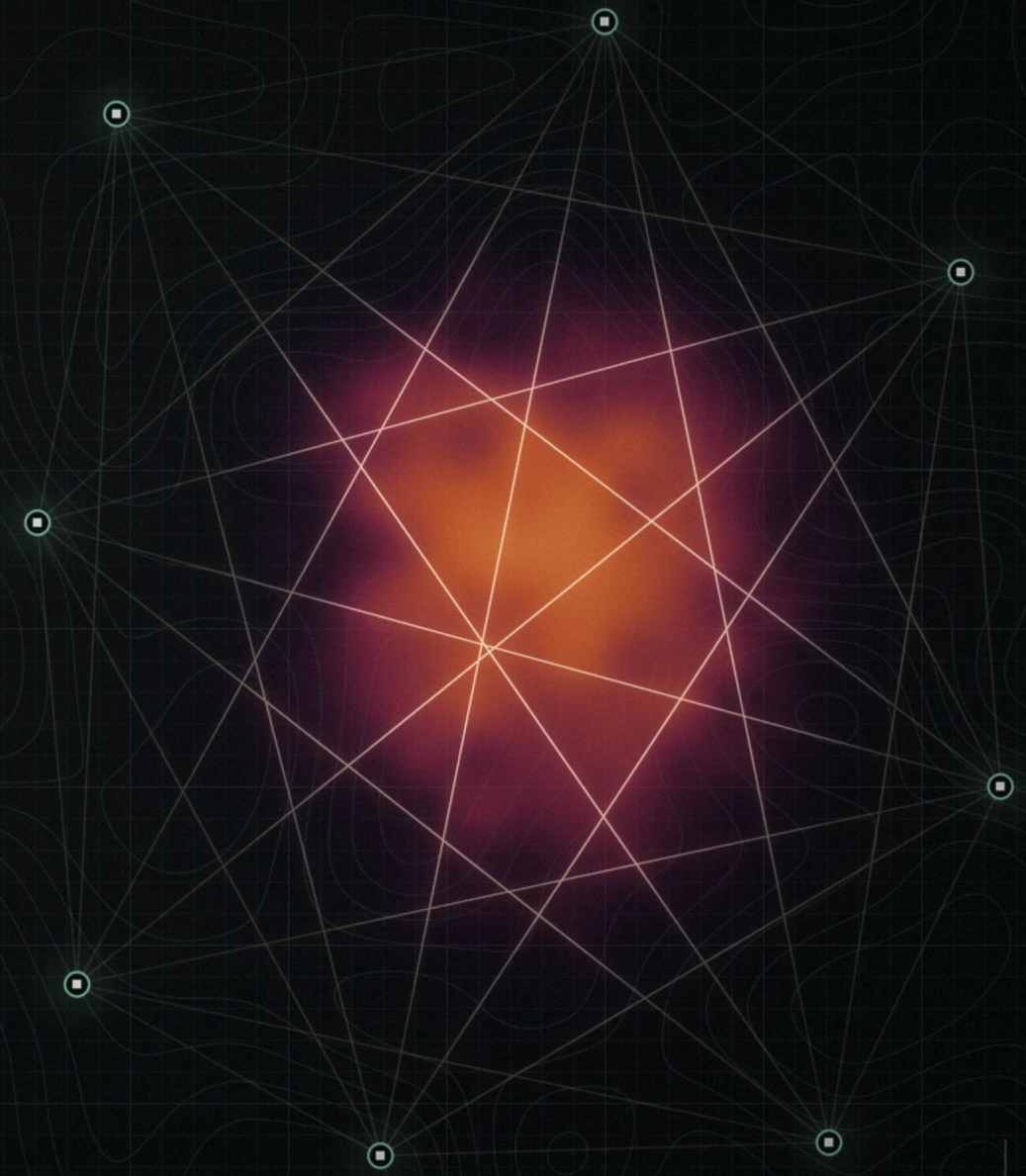
01	**For the team at the door**	Tired, loud, rushed: it must read at a glance
02	**Nothing on the person**	No tag or beacon to hack, intercept or lose
03	**Placed from outside**	No one goes in to set it up
04	**Room-level is enough**	Operators trade precision for less to carry
05	**No network on site**	Pucks bring their own links: no GPS, cell or cloud

> These are limits of the current design, not the approach.
> Future versions can be made far better: finer positions, people holding still, more wall types.

THE INSIGHT

A moving body casts a radio shadow.

Every WiFi link between two pucks flickers when someone moves through it. Catch that shadow on every link, and you can map where the movement is.



Five steps from radio shadows to a live map

01

Pucks

ESP32-C5 boards trade small WiFi frames

→

02

Multi-static links

Every sender-to-receiver path is a link: 4 today

→

03

Motion score

How much each link's signal flickers

04

Heat map

Evidence spread along each link's path

→

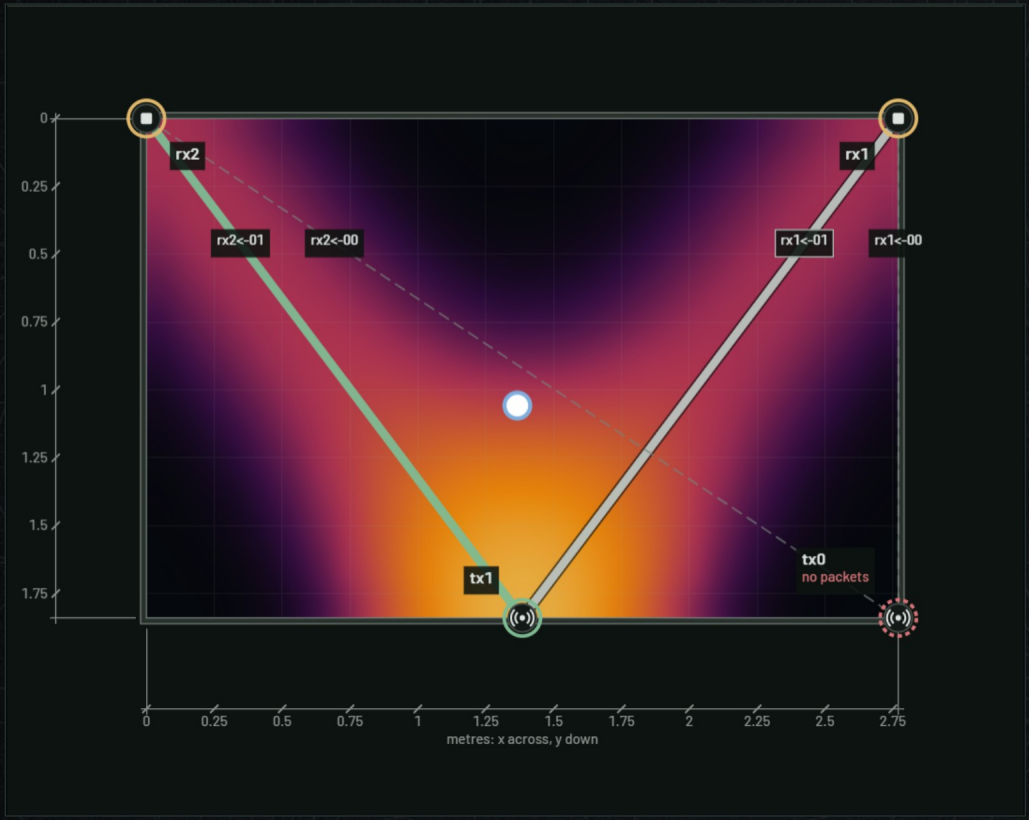
05

Dashboard

Live map and link health in a browser

114 subcarriers per packet · about 1 Hz high-pass over a 2 s window · radio tomographic imaging at 6 or more links

The MVP maps movement live



4 pucks

2 senders and 2 receivers, around the room

4 links

Every receiver hears every sender

114

Subcarriers measured on every packet

Offline

One laptop and a browser. No cloud.

Live capture with one sender offline: the map keeps running on the links still live.

The situation changes. Shadowcast adapts.

01 · EVACUATION

Who is still inside?

Pucks around the building show which rooms still have movement, and which are clear.

→

02 · COLLAPSE

The map keeps going

Lost pucks drop only their own links. The alarm re-scales to the links still live.

→

03 · RESCUE

Find who can’t move

New pucks go in by hand or drone. Breathing mode and probe pucks search the rubble.

TODAY

The alarm counts only live links, and baselines self-adjust when a room can’t be cleared.

BUILT OUT

Breathing mode, probe pucks, drone re-drops and pucks that locate themselves.

Pucks go around a room in minutes, with no line of sight

	Handheld through-wall radar	Shadowcast
Placement	Pressed against one wall by an operator	Pucks around the area, then walk away
What you get	About a minute per reading, from one wall	A live map across every link at once
Still people	Detects breathing, up to about 50 ft	Not yet: moving people only (roadmap)
Cost	About $6,000 per unit	About $15 per ESP32-C5 dev board
To cover more	More units, more operators	More pucks, and the links multiply

Radar figures from public reporting on the L-3 Range-R. Board price is Espressif's ESP32-C5-DevKitC-1 at launch; batteries and cases extra.

SCALES TO A BUILDING

Every puck you add multiplies the links

30
links from 6 pucks

240
links from 16 pucks

Once every puck both sends and receives, N pucks give N × (N – 1) links. About 16 pucks cover half of this floor.

Plan of our BASE-X floor 14 model: 16 pucks around the north half, every pair linked, two people moving.

From this floor to the field

	Today, on this floor	Built out
Sensing	Walking people, through drywall	Still people by breathing; rubble voids
Placement	By hand, in minutes	Rugged 1.5 in pucks by drone, self-locating
Signal	Our own pucks, about 1% of airtime	The building's own WiFi (exploring now)
Power	Pucks on battery, receiver on USB	All on battery: 7 to 11 h per 1,000 mAh cell
Output	Heat map plus a yes/no room alarm	Phone app and ATAK; red/green light on pucks
Security	No beacon on the person to hack or track	Encrypted links; WiFi riding is listen-only
Reliability	0 false alarms on held-out recordings	Lose 1 of 16 pucks, keep 210 of 240 links

Battery estimate: ESP32-C5 receive current of 89 to 135 mA (Espressif, via DigiKey). ATAK markers would carry their error radius.

Shadowcast at a glance

OBJ.01

DUAL USE

The mission

Entry teams and rescuers see who is moving behind walls. Nobody carries a tag.

4 links

LIVE ON THIS FLOOR

Working today

Four pucks and a receiver, built this weekend. You run the blind trials.

$15

PER BOARD

Ready for the field

Offline, no cloud, no network needed. Built out: battery pucks and a phone app.

N(N–1)

LINKS FROM N PUCKS

The idea

Multi-static radio shadows: every sender and receiver pair is its own sensor.

SHADOWCAST · TEAM 12

Never go in blind.

Thank you. Questions?

Nicolas · Simon · Andrew · Siv

github.com/CodeMetalForge-Hackathon/Team-12

It degrades, it doesn't die

	Sensing	What the operator sees
Primary	A ring of pucks, our own links	Heat map in the phone app or ATAK
Alternate	The building's own WiFi	Yes/no room alarm on any screen
Contingency	Lose pucks, keep the other links	Red/green light on each puck
Emergency	One puck pair across a doorway	One light: moving inside, or not

Running today: the puck ring, the heat map and the room alarm. The phone app, ATAK, WiFi riding and puck lights are built out.

Walk in, and the heat follows

01

Every link measured

Loss, packets per second, RSSI and clipping for every link

02

Someone walks in

The heat map follows them across the room

03

Blind trials

A judge draws a position or ‘none’ from a card. We keep score.

04

The numbers

Accuracy and false alarms from our evaluation tool

Four pucks around a BASE-X conference room: two senders, two receivers and four live links.

Adding pucks made it worse, so we traced why

What broke	Evidence	What we changed
Each receiver prints only about 100 packets/s, shared by all its senders	CSI goes out as ~0.9 KB of text per packet: 9.5 ms a line over serial	Per-packet debug line removed; sender rate set by sender count
Receiver gain locks from a mix of senders	Any reset shifted scores 3 to 5×, silently breaking saved thresholds	Gain lock to the strongest sender; thresholds record their settings
The filter threw away most of the walking signal	68% of walking energy sits at 0.3 to 1 Hz; the high-pass was at ~2 Hz	About 1 Hz high-pass, 2 s window, median prefilter
Every added link made an empty room look busier	False blobs in 41 to 68% of empty frames; a fixed ‘any 2 links’ alarm	Coverage-normalized map; the alarm scales with link count

Detection up, false alarms down

95%

Weak-link detection on held-out recordings

Up from 80%

0

False alarms on that held-out data

With the new detector

8.5×

Strong-link margin over threshold

Up from 4.6×

4 to 6

Modeled false alarms per hour at 16 links

Down from about 72

Bench results from our own recordings. On-site BASE-X numbers come from the live blind trials.

From one room to a whole building

01	**Pucks take turns**	Every puck sends and receives: 16 pucks, 240 links
02	**WiFi tapping**	Exploring now: ride existing WiFi, not our own signal
03	**No cables**	Scores computed on each puck, relayed to a phone app
04	**Self-locating pucks**	Pucks range each other and map themselves in 5 min
05	**Drone-dropped pucks**	Rugged 1.5 in pucks, placed by hand or by drone
06	**Still people and breathing**	Dual-antenna pucks; probe pucks for rubble voids
07	**Building scale**	6 to 8 per house floor, 10 to 12 per small building

What it does today, and what it does built out

TODAY, LIVE ON THIS FLOOR

- ✓ Walking people
- ✓ Their rough area, within a few meters
- ✓ Through drywall
- ✓ Measured on site, in blind trials

BUILT OUT

- People standing still, by their breathing
- Voids in rubble, with probe pucks
- Riding the building’s own WiFi
- Rugged pucks, dropped by drone

The physics limit at 2.4 GHz is heavy reinforced concrete: two such walls cost about 75 dB.

How the room decision avoids false alarms

The room reads MOTION only when	Why
About 15% of live links agree, and at least 2	Scales with link count, unlike a fixed ‘any 2 links’
They come from at least 2 receivers and 2 senders	One misbehaving puck can’t raise the alarm alone
They cross the same area	A real person disturbs the links where they cross
It holds for 0.5 s	Brief spikes don’t count
It isn’t every link on one puck at once	That pattern is a hardware glitch, and it’s ignored

Modeled false alarms at 16 links: about 72 an hour before, 4 to 6 after.

Where the pucks go sets the accuracy

Surround the area

Pucks on at least 3 sides, ideally 4, so links cross the room. Pucks on the same wall add almost nothing.

Spacing

Every 2 m gives about 1 to 1.5 m accuracy; every 3 m, room level. Published sub-meter systems used about one node per meter.

Mounting

Chest height (1.0 to 1.2 m), rigid, same orientation, at least 0.5 m from metal.

Range

About 15 m of radio range is plenty. A link mostly sees motion within 1 to 2 m of its line.

With 6 pucks, an 18 × 15 m area is about 90% resolved when every puck sends and receives, and about 50% with fixed roles.

Built on published through-wall sensing work

Work	What it showed
Wilson & Patwari, radio tomographic imaging (2010, 2011)	34 nodes around part of a house tracked people through brick to 0.45 to 1 m. Commercialized by Xandem.
Maas et al. (2013)	Under 2 m accuracy with about 10 radios per floor
Kaltiokallio, Bocca & Patwari	Channel diversity, multi-target tracking, the Spin token-passing protocol
Zhao & Patwari	Kernel-distance RTI, which also finds people standing still given an empty baseline
Daqing Zhang's group	Fresnel-zone models; FarSense, breathing at 8 to 9 m from two antennas
Espressif esp-csi, ESPARGOS, IEEE 802.11bf	CSI tooling we build on, phase-coherent ESP32 arrays, the WLAN sensing standard